TP1 - Variable Types & Descriptive Statistics

Exploratory Data Analysis & Unsuperivsed Learning
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Objective: This initial practical session is designed to enhance your understanding of various data variable types and the corresponding statistical and graphical tools suitable for each type.

> **The `Jupyter Notebook` for this TP can be downloaded here:** TP1-Gapminder.

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1. `Gapminder` dataset

`Gapminder` is an excerpt of data available at https://gapminder.org. For each of 142 countries (`country`), the package provides values for life expectancy (`lifeExp`), GDP per capita (`gdpPercap`), and population (`pop`), every five years, from 1952 to 2007 (`year`). It is initially used in Jennifer Bryan's excellent `gapminder` teaching package for R (`ggplot`, `tidyverse`, and more). For more information about `gapminder`: - Documentation: https://www.gapminder.org/data/documentation/ - A short demonstration video is available here: Hans Rosling's 200 Countries, 200 Years, 4 Minutes - The Joy of Stats - BBC Four.

```
# useful packages
import numpy as np
import pandas as pd

#%pip install gapminder  (for installing gapminder if you haven't had one)
from gapminder import gapminder
print(f"* Number of observations: {gapminder.shape[0]}")
gapminder.sample(3)
```

```
* Number of observations: 1704
```

	country	continent	year	lifeExp	pop	gdpPercap
1528	Thailand	Asia	1972	60.405	39276153	1524.358936
998	Mongolia	Asia	1962	48.251	1010280	1056.353958
530	France	Europe	1962	70.510	47124000	10560.485530

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2. Variable types

`EDA` involves summarizing and visualizing data to uncover patterns, detect anomalies, and understand relationships between variables. Statistical summaries, such as mean, median, and standard deviation, are essential tools in this process.

- Which variables are considered quantitative and which are qualitative?

> Hint: You can check the default column types by using `gapminder.dtypes`. But this may not be accurate because some categorical data may be encoded using numerical values.

```
# TODO: variable types
```

> **Your response**:

2.1 Quantitative variables

1. In year 2002,

- Compute suitable statistics for each quantitative variable (excluding `year`) to obtain an overall summary.
- Recall the definitions of **skewness** and **kurtosis** introduced in the course.
- Compute these metrics for each quantitative variable and explain the distribution of each variable based on these values.

```
# TODO
```

2. Graphically represent the distribution of each variable for the year 2002. After plotting the distributions, provide a brief explanation for each variable.

```
# TODO:
```

3. According to the data, in 2002:

- Which country is the richest?
- Which country is the poorest?
- Which country is the healthiest?
- Which country is the unhealthiest?

```
# TODO:
```

4. Repeat the previous question for the year 1977. Before computing, can you guess which country had the lowest life expectancy around that year?

```
# TODO:
```

2.2. Qualitative variables

Qualitative variables are simpler than the quantitative ones, as we primarily focus on the proportion or frequency of each category. In our dataset, the existing qualitative variables are not suitable for analysis because they are repeated each year. Therefore, we will create three new qualitative variables associated with the three quantitative columns by copping them into three categories each.

In year 2002,

1. Add the following three variables to the `gapminder` dataset by grouping each quantitative variable into 3 groups.

- Create variable `gdpQual` with three categories: [**"developing"**, **"moderate"**, **"developed"**] using variable `gdpPercap`.
- Create variable `popQual` with three categories: [**"small"**, **"medium"**, **"large"**] using variable `pop`.
- Create variable `lifeExpQual` with three categories: [**"unhealthy"**, **"moderate"**, **"healthy"**] using variable `lifeExp`.

> Hint: you may find `pd.cut` function helpful.

```
# TODO:
```

2. Compute the appropriate statistical values and graphically represent the distribution of each newly created qualitative variable.

```
# TODO
```

- Graphically represent them and describe the graph.

```
# TODO
```

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3. Time evolution

`Gapminder` captures global changes from 1952 to 2007. It is more insightful to examine how these variables evolve over time.

3.1. Evolution of quantitative columns

1. Create a line plot of `lifeExp` for the five continents from 1952 to 2007 using `sns.lineplot`. What observations can you make from the plot?

```
# To do
```

2. Repeat the same process with `pop`, and then with `gdpPercap`. Provide your comments for each case.

```
# To do
```

3. Plot the evolution of each of the three quantitative variables for **Cambodia** vs **Thailand**. Describe the graphs.

```
# To do
```

3.2. Evolution of qualitative columns

1. Visualize the evolution of `lifeExpQual` column in **Asia** from 1952 to 2007. What do you observe?

```
# To do
```

2. Do the same with `popQual` and `dgpQual` columns.

```
# To do
```

3. Visualize the evolution of the three qualitative columns above for African countries from 1952 to 2007.

Further readings

- Gapminder documentation: https://www.gapminder.org/data/documentation/
- A short demonstration video is available here: Hans Rosling's 200 Countries, 200 Years, 4 Minutes - The Joy of Stats - BBC Four.
- Graphical tools:
  - `matplotlib`
  - `seaborn`
  - `https://plotly.com/python/`